TOPS Technologies

Interview Questions — DA- Working with NumPy (python)

Total Questions: 47

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1. What is NumPy and why is it popular among data analysts?

> **Hint:** Think about performance, large datasets, and mathematical operations.

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2. Explain the main difference between a Python list and a NumPy array.

> **Hint:** Consider how data is stored and processed in each.

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3. What is vectorization in NumPy, and why is it important for data analysis?

> **Hint:** How does it affect the speed of operations on large datasets?

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4. How would you compare the performance of a Python list and a NumPy array when performing element-wise addition on large datasets?

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5. How do you create a NumPy array from a regular Python list?

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6. Write code to create a NumPy array containing numbers from 0 to 9 (inclusive).

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7. What is the difference between np.arange() and np.linspace() when creating arrays?

> **Hint:** Think about how the spacing between elements is defined.

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8. How would you create an array of 5 equally spaced values between 1 and 10 using NumPy?

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9. Which function would you use to create an array filled with zeros? What about an array filled with ones?

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10. Write code to generate a 3x3 identity matrix using NumPy.

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11. How would you generate a 1D array of 10 random integers between 1 and 100 using NumPy?

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12. What is the purpose of np.random.seed()? When should you use it?

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13. How can you check the number of dimensions and the shape of a NumPy array?

> **Hint:** Look for attributes that describe the array's structure.

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14. If you have a NumPy array of shape (4, 3), how would you reshape it to (2, 6)?

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15. What is the difference between ravel(), flatten(), and resize() in NumPy?

> **Hint:** Consider whether the result is a copy or a view, and if the original array is affected.

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16. How do you transpose a 2D NumPy array?

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17. Show how to perform element-wise addition and multiplication between two NumPy arrays of the same shape.

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18. What is broadcasting in NumPy? Give a simple example where it is useful.

> **Hint:** Think about operations between arrays of different shapes.

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19. Explain the difference between element-wise operations and aggregate operations in NumPy.

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20. How would you use NumPy to check which elements of an array are greater than 10?

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21. Write code to calculate the square root of every element in a NumPy array.

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22. How do you find the mean and standard deviation of a NumPy array?

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23. What is the difference between np.sum() and np.cumsum()?

> **Hint:** Think about the type of result each function returns.

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24. How would you round all values in a NumPy array to two decimal places?

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25. Write code to find the maximum value and its index in a NumPy array.

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26. How do you access the third element of a 1D NumPy array?

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27. Given a 2D array, how do you select all elements from the second row?

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28. Explain how negative indexing works in NumPy arrays.

> **Hint:** Focus on what negative indices refer to.

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29. Write code to select every second element from a NumPy array using slicing.

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30. What is fancy indexing in NumPy? Provide a simple example.

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31. How would you use boolean indexing to select all even numbers from a NumPy array?

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32. Describe how you would use np.where() to find the indices of all elements in an array that are less than 5.

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33. How do you concatenate two 1D arrays horizontally using NumPy?

> **Hint:** Which function joins arrays along the existing axis?

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34. What is the difference between stacking and concatenation in NumPy?

> **Hint:** Consider how the dimensions of the result may change.

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35. Write code to split a 1D NumPy array into three equal parts.

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36. How can you insert a new value at a specific position in a NumPy array?

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37. Explain the difference between a copy and a view in NumPy. Why does it matter?

> **Hint:** Think about what happens when you modify the new array.

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38. How would you find all unique values in a NumPy array?

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39. What does np.argsort() return and how can it be useful?

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40. Write code to find the indices of all non-zero elements in a NumPy array.

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41. How do you check if any or all elements of an array meet a certain condition using NumPy?

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42. What is the purpose of np.isnan(), np.isinf(), and np.isfinite()?

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43. How would you replace all NaN values in an array with the mean of the non-NaN values?

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44. Describe a step-by-step approach to cleaning and analyzing a small dataset using NumPy arrays (for example, prices and quantities).

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45. If you have arrays of prices and quantities, how would you calculate the total value for each product using NumPy?

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46. How could you use an AI tool like ChatGPT or GitHub Copilot to help you write or debug NumPy code? What would you do to make sure the code suggestions are safe and correct?

> **Hint:** Think about verifying results, understanding code, and not copying blindly.

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47. Imagine you have a large dataset with missing values. How might AI-powered tools help you handle and clean this data using NumPy, and what would you check before trusting their output?
